

Yash Mehrotra . Jarvis Consulting

Software developer with 4+ years of professional software engineering experience and a graduate certificate in Artificial Intelligence with Machine Learning from Humber College. Experienced in building production applications, backend systems, data pipelines, APIs, and user-facing features using C#, Python, SQL, cloud technologies, and modern development frameworks. Worked with frontend technologies including React, Vue, and Flutter, providing a strong foundation in component-based UI development, state management, routing, API integration, and application architecture. Comfortable working across technical and non-technical teams, collaborating in Agile environments, and quickly learning new technologies. Particularly interested in building reliable, maintainable enterprise software and full-stack applications that solve practical business problems.

Skills

Proficient: C#, Python, RDBMS/SQL, Agile/Scrum, Git, Linux/Bash

Competent: React, Vue, Flutter, Docker, Databricks, PostgreSQL, REST API

Familiar: Azure, Spark, ADLS, Azure Data Factory, Angular

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_YashMehrotra

Cluster Monitor [GitHub]: Developed a Linux Cluster Monitoring Agent using Bash scripting, PostgreSQL, Docker, and Crontab to automatically collect and store hardware specifications and real-time resource usage across multiple cluster nodes. Designed two core scripts, one to capture static host information at provisioning time and another to record dynamic CPU and memory usage on a scheduled interval, with all data persisted to a PostgreSQL instance running in a Docker container. Enabled Linux administrators to query historical usage data for capacity planning and failure detection across distributed node environments.

Online Sales Dataset Analysis and Restock Recommendation System [GitHub]: Developed a Python and SQL-based analytics workflow for a 50,000+ transaction Online Sales Dataset to uncover insights around seasonal demand patterns, discount effectiveness, return behaviour, and customer segmentation via RFM scoring. Built reusable data-processing workflows using Pandas and trained a Facebook Prophet time-series forecasting model to predict monthly product demand, with dynamic safety buffers derived from each product's historical demand volatility. Delivered a reusable Python inference function enabling per-product restock recommendations for any target month on demand.

Databricks Data Pipeline [GitHub]: Developed scalable ETL and analytics pipelines using PySpark, Apache Spark, Databricks, SQL, and Delta Lake to ingest and transform data from CSV, Parquet, Azure SQL, cloud storage, and external APIs. Implemented a Medallion Architecture with Bronze, Silver, and Gold layers for raw ingestion, data cleansing, validation, enrichment, joins, and business-ready aggregations. Integrated Azure services including ADLS, Azure Data Factory, JDBC, Lakeflow Connect, and Unity Catalog, and built Delta Live Tables pipelines and interactive Databricks dashboards for fraud and stock-market analytics.

Home Credit Default Risk Prediction [GitHub]: Developed an end-to-end credit risk application in Databricks using 300,000+ Home Credit loan applications, implementing a Bronze–Silver–Gold medallion architecture for ingestion, preprocessing, feature engineering, and model-ready datasets with Spark, SQL, and Pandas. Built and optimized Logistic Regression, Random Forest, and Gradient Boosting models using Scikit-learn and cross-validation, achieving ~0.76 AUROC and evaluating performance across Gini, KS, AUPRC, F1, and calibration metrics. Created Databricks EDA and model-monitoring dashboards and a Streamlit application for real-time applicant scoring and explainability.

Stock Market Deep Learning System [GitHub]: Developed an end-to-end deep learning and backtesting pipeline for US equity return forecasting across 50 stocks from 2005–2025. Built and compared feedforward neural networks, LSTM sequence models, linear regression, and ResNet-18 chart-image models using PyTorch, Scikit-learn, Pandas, and time-series walk-forward validation with strict leakage controls. Engineered technical and market features including returns, implemented reproducible chronological preprocessing, scaling, checkpointing, and inference workflows. Evaluated models using Sharpe ratio, drawdown, turnover, and transaction-cost-adjusted portfolio performance, with the selected feedforward model achieving 52.08% directional accuracy and a 1.17 net Sharpe ratio. Built a realistic backtesting framework with SPY and equal-weight benchmarks, cost sensitivity analysis, and monitoring.

Highlighted Projects

Codebase RAG Assistant [GitHub]: Built a Source-Grounded Codebase Assistant using ChromaDB, Ollama, and a custom RAG pipeline to answer natural language questions about large C++ codebases with strict citation enforcement. Implemented a custom C++ code chunker using sentence transformer embeddings and vector indexing to enable semantically accurate retrieval across complex codebases, exposed via an MCP server interface. Designed the system to ground every response in retrieved source chunks, preventing hallucination and ensuring all answers are traceable back to the original codebase.

Domain Intelligence Agent [GitHub]: Built a modular Agentic System using Python, FastAPI, and Streamlit to analyze financial data through a shared core architecture of specialized planning, retrieval, analysis, and reporting agents. Designed a domain-agnostic orchestration layer that routes queries through the appropriate pipeline, grounding outputs in real-time public financial data via Yahoo Finance. Delivered an interactive Streamlit interface for querying and reporting across the agent pipeline.

Professional Experiences

ML Engineer, Jarvis (2026-present): Developed enterprise-style software, data engineering, and machine learning solutions using Python, SQL, PySpark, Databricks, PostgreSQL, Docker, Bash, and Azure technologies. Built reusable ETL pipelines, database-backed applications, infrastructure monitoring utilities, REST-oriented integrations, analytics workflows, and predictive systems. Implemented Medallion Architecture pipelines with Azure Data Lake Storage, Azure Data Factory, JDBC, Delta Lake, and Databricks while applying Git-based version control, modular development practices, technical documentation, testing, and Agile development methodologies.

Game Developer, Hashbyte Studio (2021-2023): Developed an end-to-end deep learning and backtesting system for US equity return forecasting across 50 stocks using PyTorch, Scikit-learn, Pandas, and walk-forward validation. Built and evaluated feedforward, LSTM, linear, and ResNet-18 models, with transaction-cost-adjusted portfolio backtesting against SPY and equal-weight benchmarks. Deployed the selected model on AWS using S3, SageMaker, Lambda, Step Functions, EventBridge, IAM, and CloudWatch for scalable inference, orchestration, and monitoring.

Education

Humber College (2025), Diploma, Artificial Intelligence with Machine Learning - GPA: 84.97/100.00

Jaypee University of Information Technology (2017-2021), Bachelor of Technology, Computer Science & Engineering

Miscellaneous

- Databricks Certified Data Engineer Associate (2026)
- Coursera|Stanford Machine Learning (2020)
- Reading sci-fi/fantasy books
- Writing videogame mods